Problem Statement vs. Key Findings Summary

Problem Statement	Key Findings
How can Airbnb hosts optimize their listing prices to increase bookings and revenue?	Manhattan listings have the highest prices but also lower review counts, indicating premium demand but lower booking churn. Brooklyn offers a better balance between affordability and occupancy.
Which areas in NYC are profitable for hosting and worth expanding?	Top concentrations of listings found in Manhattan & Brooklyn, especially Harlem, Midtown, and Williamsburg — hotspots for profitability.
Do room types significantly affect pricing and guest demand?	The entire home/apartment commands the highest prices and is the strongest feature in predicting Airbnb price. Private rooms show high demand at lower prices.
How does availability influence performance and listings category?	Listings with extremely high yearly availability often show lower demand or overpriced structure — strong opportunity to optimize pricing.
Can guests be segmented to improve pricing strategy?	K-Means clustering identified 4 customer listing segments: (1) Luxury Low-Demand, (2) Budget High-Demand, (3) Mid-priced Fully Booked, (4) Mid-priced Low Demand — each with tailored pricing recommendations.
Which features most strongly influence price predictions?	Feature importance reveals room type, location coordinates (latitude/longitude), and availability_365 as major predictors for pricing.

Airbnb Analysis Report

Prepared By Soniya

Client / Stakeholder : Airbnb Company

Purpose :

This project will analyze the Airbnb dataset with a focus on price analysis. The analysis will include examining trends, identifying key factors influencing room types, price, availability, number of reviews and segmenting insights by location. Based on the findings, the project will suggest optimal pricing strategies.

Summary :

This report provides a comprehensive analysis of over 46,000 Airbnb listings in New York City to uncover trends in pricing, guest behavior, and market performance. Insights reveal that Manhattan and Brooklyn dominate the market with the highest pricing and listing demand, while Entire home/apartment rentals remain the most preferred accommodation type among travelers. Despite price differences, listings across all boroughs show strong engagement with over 1 million total reviews and an average occupancy rate of around 70%. The findings highlight key opportunities for hosts to optimize pricing strategies, improve listing quality, and expand in high-demand neighborhoods to maximize revenue and guest satisfaction.

Phase I - Ask :

- Identified Business task
- Considered key Stakeholders
- BrainStormed SMART questions

❖ Business Task :

- Airbnb, a home-stay company in NYC, aims to understand how other factors influence prices. The goal of this analysis is to identify patterns that can help find factors that influence the prices and build a model that predicts optimal prices for new or existing listings.

❖ Stakeholder:

- Mr. X (Marketing Director)

***Note**: SMART questions sheet is attached separately

Phase II - Prepare :

- The dataset includes Airbnb (NYC) data.
- Each record represents a room stay with details such as id, name, host id, host name, neighbourhood group, neighbourhood, latitude, longitude, room type, price, minimum nights, number of reviews, last review, reviews per month, calculated host listings count, availability 365.
- Got a dataset from Kaggle.
- The dataset is relevant, credible, and historical data.
- The data is well organized in row, column manner in **Google Sheets**.
- Columns used for analysis are room type, neighbourhood group, price, number of reviews, availability 365, latitude, longitude, minimum nights.

Phase III - Process :

- ➢ **Tool Used** : Kaggle Notebook, Python
- ➢ Ensured Data Integrity.
- ➢ Calculate trip duration in minutes,weekday and month from date - time columns and age from birth year.
- ➢ Some records had unrealistic prices, which were treated as errors and removed.
- ➢ Removed null or blank values in name, host name, last review, reviews per month.
- ➢ Removed Duplicate records.
- ➢ Drop rows where ‘name’ or ‘host_name’ are missing.
- ➢ Filling missing values in ‘last review’ and ‘reviews per month’.
- ➢ Removed records whose price is lesser than 30 to avoid outliers.
- ➢ Verified data types and formatting for consistency.
- ➢ The data is cleaned, organized and well maintained.

```python
# Step 1: Import libraries
import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Step 2: Load dataset
df = pd.read_csv('/kaggle/input/new-york-city-airbnb-open-data/AB_NYC_2019.csv')

# Step 3: Check first few rows
print("rows, cols:", df.shape)
df.head()
```

Fig 1.0

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80

Fig 1.1

```python
# 1. Overview checks
df.info()
print("\nMissing values per column:\n", df.isnull().sum())
print("\nDuplicate rows:", df.duplicated().sum())
display(df.describe(include='all').T)
```

Fig 2.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48895 entries, 0 to 48894
Data columns (total 16 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   id                              48895 non-null  int64
 1   name                            48879 non-null  object
 2   host_id                         48895 non-null  int64
 3   host_name                       48874 non-null  object
 4   neighbourhood_group             48895 non-null  object
 5   neighbourhood                   48895 non-null  object
 6   latitude                        48895 non-null  float64
 7   longitude                       48895 non-null  float64
 8   room_type                       48895 non-null  object
 9   price                           48895 non-null  int64
 10  minimum_nights                  48895 non-null  int64
 11  number_of_reviews               48895 non-null  int64
 12  last_review                     38843 non-null  object
 13  reviews_per_month               38843 non-null  float64
 14  calculated_host_listings_count  48895 non-null  int64
 15  availability_365                48895 non-null  int64
dtypes: float64(3), int64(7), object(6)
```

Fig 2.1

```
Missing values per column:
 id                                  0
name                                16
host_id                              0
host_name                           21
neighbourhood_group                  0
neighbourhood                        0
latitude                             0
longitude                            0
room_type                            0
price                                0
minimum_nights                       0
number_of_reviews                    0
last_review                      10052
reviews_per_month                10052
calculated_host_listings_count       0
availability_365                     0
dtype: int64

Duplicate rows: 0
```

Fig 2.2

	count	unique	top	freq	mean	std	min	25%
id	48895.0	NaN	NaN	NaN	19017143.23618	10983108.38561	2539.0	9471945.0
name	48879	47905	Hillside Hotel	18	NaN	NaN	NaN	NaN
host_id	48895.0	NaN	NaN	NaN	67620010.64661	78610967.032667	2438.0	7822033.0
host_name	48874	11452	Michael	417	NaN	NaN	NaN	NaN
neighbourhood_group	48895	5	Manhattan	21661	NaN	NaN	NaN	NaN
neighbourhood	48895	221	Williamsburg	3920	NaN	NaN	NaN	NaN
latitude	48895.0	NaN	NaN	NaN	40.728949	0.05453	40.49979	40.6901
longitude	48895.0	NaN	NaN	NaN	-73.95217	0.046157	-74.24442	-73.98307
room_type	48895	3	Entire home/apt	25409	NaN	NaN	NaN	NaN
price	48895.0	NaN	NaN	NaN	152.720687	240.15417	0.0	69.0
minimum_nights	48895.0	NaN	NaN	NaN	7.029962	20.51055	1.0	1.0
number_of_reviews	48895.0	NaN	NaN	NaN	23.274466	44.550582	0.0	1.0
last_review	38843	1764	2019-06-23	1413	NaN	NaN	NaN	NaN
reviews_per_month	38843.0	NaN	NaN	NaN	1.373221	1.680442	0.01	0.19
calculated_host_listings_count	48895.0	NaN	NaN	NaN	7.143982	32.952519	1.0	1.0
availability_365	48895.0	NaN	NaN	NaN	112.781327	131.622289	0.0	0.0

Fig 2.3

```python
# Drop rows where 'name' or 'host_name' are missing
df = df.dropna(subset=['name', 'host_name'])

# Fill missing values in 'last_review' and 'reviews_per_month'
df['last_review'] = df['last_review'].fillna('No Review')
df['reviews_per_month'] = df['reviews_per_month'].fillna(0)

# Check again
df.isnull().sum()
```

Fig 3.0

```python
df['last_review'] = pd.to_datetime(df['last_review'], errors='coerce')
print(df.last_review)
```

Fig 4.0

```
Out[3]:
id                                0
name                              0
host_id                           0
host_name                         0
neighbourhood_group               0
neighbourhood                     0
latitude                          0
longitude                         0
room_type                         0
price                             0
minimum_nights                    0
number_of_reviews                 0
last_review                       0
reviews_per_month                 0
calculated_host_listings_count    0
availability_365                  0
dtype: int64
```

Fig 3.1

```
0       2018-10-19
1       2019-05-21
2              NaT
3       2019-07-05
4       2018-11-19
           ...
48890          NaT
48891          NaT
48892          NaT
48893          NaT
48894          NaT
Name: last_review, Length: 48858, dtype: datetime64[ns]
```

Fig 4.1

Phase IV - Analyse :

Fig 5.0:

- ➢ It provides descriptive statistics of a Dataframe.
- ➢ It provides mean,count,min, percentiles, max and standard deviation.
- ➢ The dataset shows **significant outliers** especially in **price** and **minimum_nights**.
- ➢ **Skewed distributions** in price and reviews indicate the need for normalization or transformation during modeling.
- ➢ **Missing review dates** must be handled either through imputation or classification as "no reviews".
- ➢ Geographic attributes confirm the dataset is focused on **NYC Airbnb properties**.

```
In [5]: df.describe()
```

Out[5]:

	id	host_id	latitude	longitude	price	minimum_nights	number_of_reviews	last_reviev
count	4.885800e+04	4.885800e+04	48858.000000	48858.000000	48858.000000	48858.000000	48858.000000	38821
mean	1.902335e+07	6.763169e+07	40.728941	-73.952170	152.740309	7.012444	23.273098	2018-10-0 07:23:47.0
min	2.539000e+03	2.438000e+03	40.499790	-74.244420	0.000000	1.000000	0.000000	2011-03-2 00:00:00
25%	9.475980e+06	7.818669e+06	40.690090	-73.983070	69.000000	1.000000	1.000000	2018-07-0 00:00:00
50%	1.969114e+07	3.079133e+07	40.723070	-73.955680	106.000000	3.000000	5.000000	2019-05-1 00:00:00
75%	2.915765e+07	1.074344e+08	40.763107	-73.936280	175.000000	5.000000	24.000000	2019-06-2 00:00:00
max	3.648724e+07	2.743213e+08	40.913060	-73.712990	10000.000000	1250.000000	629.000000	2019-07-0 00:00:00
std	1.098289e+07	7.862389e+07	0.054528	0.046159	240.232386	20.019757	44.549898	NaN

Fig 5.0

Fig 6.0:

- ➢ 50% of the listings fall within the price range of **₹69–₹175** (or $69–$175).
- ➢ Since the IQR value is **106**, it indicates a **moderate spread** of prices among most listings.
- ➢ Anything significantly below Q1 or above Q3 (beyond determined limits later) will be treated as **potential outliers**.

```
In [6]:
# Step 1: Calculate Q1, Q3, and IQR for 'price'
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

Q1: 69.0
Q3: 175.0
IQR: 106.0
```

Fig 6.0

Fig 7.0:

- ➢ The lower limit is **-90**, which is not realistic for price values.
- ➢ Therefore, any value **below 0** can be considered invalid.
- ➢ The upper limit is **334**, meaning:
- ➢ Listings priced **above ₹334/$334** are considered **outliers**.
- ➢ These high-end properties can **skew the distribution** and may need to be capped or removed depending on analysis goals.

```
In [7]:
# Step 2: Define limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

Lower Limit: -90.0
Upper Limit: 334.0
```

Fig 7.0

Fig 8.0:

- ➢ Approximately 6% of the listings had price values that were considered outliers or unrealistic.
- ➢ Removing these extreme values helps:
  - ○ Reduce skewness
  - ○ Improve reliability of statistical analysis
  - ○ Improve performance of predictive models

```
In [8]:
# Step 3: Remove outliers
df = df[(df['price'] >= lower_limit) & (df['price'] <= upper_limit)]

print("After removing outliers, remaining rows:", df.shape)

After removing outliers, remaining rows: (45887, 16)
```

Fig 8.0

Fig 9.0:

- The **last_review** column now clearly distinguishes listings that have **never received a review**.
- The dataset is now **complete with no missing values**, making it ready for further data analysis and modeling.

```
In [9]: # Step 1: Check missing values again

        df['last_review'].fillna('No Review', inplace=True)
        df.isnull().sum()

Out[9]:
id                                0
name                              0
host_id                           0
host_name                         0
neighbourhood_group               0
neighbourhood                     0
latitude                          0
longitude                         0
room_type                         0
price                             0
minimum_nights                    0
number_of_reviews                 0
last_review                       0
reviews_per_month                 0
calculated_host_listings_count    0
availability_365                  0
dtype: int64
```

Fig 9.0

Fig 10.0 :

The distribution is **right-skewed**, meaning:

- Most listings are priced on the **lower to moderate range**.
- A smaller number of listings are in the higher price range.

The **majority of Airbnb listings** fall between:

- **$50 to $150**

The **peak density** (most common price range) is around:

- **$80 to $120**
- The long tail towards the right indicates that a few listings are **still relatively expensive** compared to the majority.

```
In [10]: sns.set(style="whitegrid", palette="pastel")

In [11]: plt.figure(figsize=(10,6))
         sns.histplot(df['price'], bins=50, kde=True)
         plt.title("Distribution of Airbnb Prices")
         plt.xlabel("Price")
         plt.ylabel("Count")
         plt.show()
```

Fig 10.0

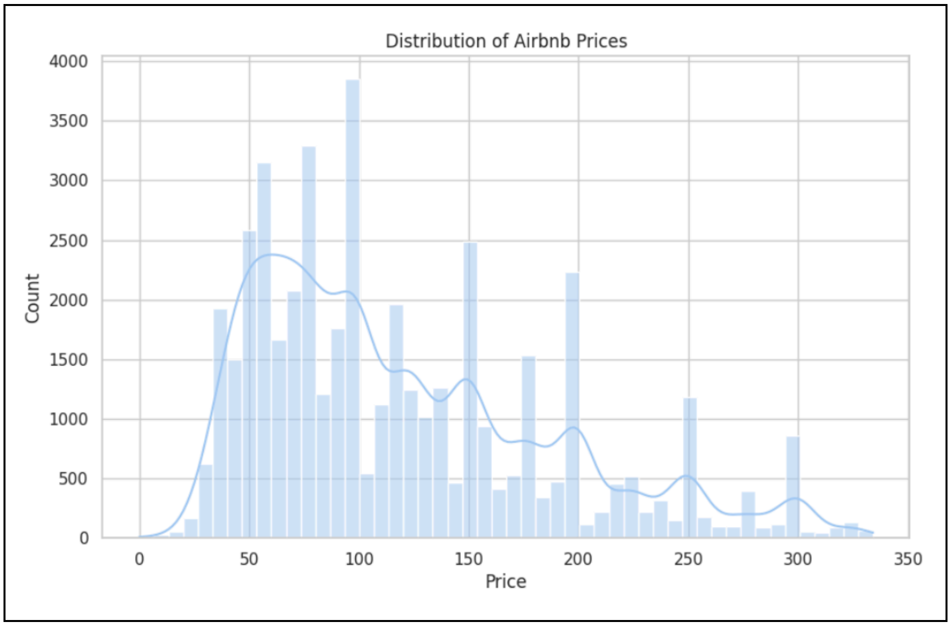

Fig 10.1

Fig 11.0 :

- ➢ Manhattan stands out as the most expensive area due to its central location, tourist attractions, and commercial hubs.
- ➢ Brooklyn also shows relatively high prices, reflecting its growing popularity.
- ➢ The Bronx and Queens provide more affordable housing options for travelers.
- ➢ Staten Island falls in the mid-range but still lower than Manhattan and Brooklyn due to lower tourism activity and accessibility.

```
In [12]:
plt.figure(figsize=(8,6))
sns.barplot(x='neighbourhood_group', y='price', data=df, estimator='mean')
plt.title("Average Price by Neighbourhood Group")
plt.xlabel("Neighbourhood Group")
plt.ylabel("Average Price")
plt.show()
```

Fig 11.0

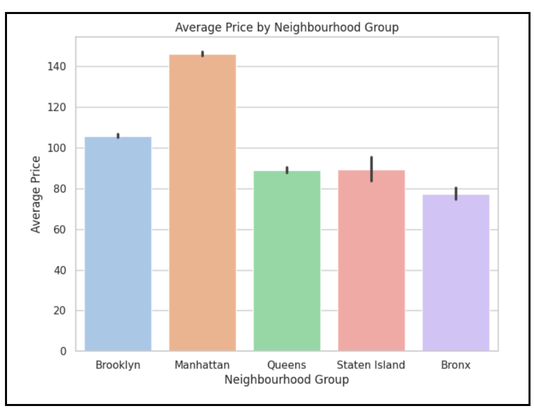

Fig 11.1

Fig 12.0 :

- ➢ **Entire homes/apartments** dominate the Airbnb market in NYC, indicating a strong preference for privacy and independent living space among visitors.
- ➢ **Private rooms** remain popular for budget-conscious travelers seeking lower costs.
- ➢ **Shared rooms** have a very limited presence, suggesting low demand for shared accommodation in a busy urban region like NYC.

```
In [13]:
plt.figure(figsize=(8,6))
sns.countplot(x='room_type', data=df)
plt.title("Room Type Distribution")
plt.xlabel("Room Type")
plt.ylabel("Count")
plt.show()
```

Fig 12.0

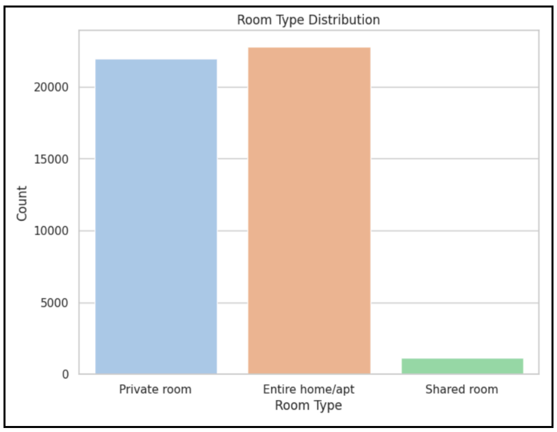

Fig 12.1

Fig 13.0 :

- ➢ The points are **widely scattered**, indicating **no clear correlation** between price and availability.
- ➢ Listings across all price ranges can be:
  - ○ Fully available (close to 365 days)
  - ○ Almost unavailable (0–20 days)
- ➢ Many listings priced **below $200** are spread across the full availability spectrum.
- ➢ Few listings are priced very low (near $0), regardless of availability — likely data anomalies or newly added listings.

```
In [14]:
plt.figure(figsize=(8,6))
sns.scatterplot(x='availability_365', y='price', data=df)
plt.title("Price vs Availability (365 Days)")
plt.xlabel("Availability (Days per Year)")
plt.ylabel("Price")
plt.show()
```

Fig 13.0

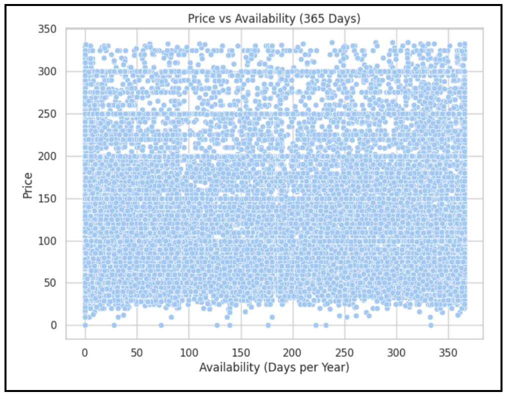

Fig 13.1

Fig 14.0 :

- ➢ There are **no strong correlations** between most numerical features.
- ➢ Price shows **very weak relationships** with reviews, availability, and minimum nights.
- ➢ The strongest useful relationship is between:
- ➢ **number_of_reviews** and **reviews_per_month** (moderate positive correlation → more popular listings get reviewed more frequently).
- ➢ Overall, **pricing and demand dynamics are not strongly explained by the numeric features alone**, suggesting other factors (like location, room type) play a bigger role.

```
In [15]:
# Select only numeric columns
numeric_df = df.select_dtypes(include=['int64', 'float64'])

# Plot correlation heatmap
plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap (Numeric Features Only)")
plt.show()
```

Fig 14.0

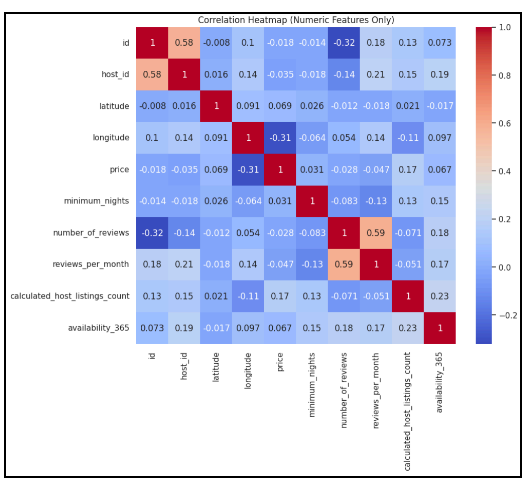

Fig 14.1

Clustering Analysis Summary:

🔍 Key Insights

➢ **Cluster 2** represents the most affordable and popular listings — private rooms with very high review counts and steady occupancy.

➢ **Cluster 1** clearly identifies luxury Manhattan apartments — high prices but low customer engagement and low bookings → potential revenue risk.

➢ **Cluster 0** hosts are earning steady revenue through consistently booked mid-range homes in Brooklyn.

➢ **Cluster 3** listings are mid-range properties in Manhattan that show weaker market traction, meaning opportunity for pricing or marketing optimization.

Out[6]:

	price	minimum_nights	number_of_reviews	reviews_per_month	calculated_host_listings_count	availability_365	latitude	longitude
cluster								
0	117.63	4.76	8.66	0.51	1.68	22.02	40.73	-73.96
1	235.95	21.79	2.15	0.67	289.17	281.96	40.73	-74.00
2	110.83	2.45	98.31	3.92	2.12	151.71	40.73	-73.94
3	126.81	14.49	14.27	0.77	10.14	286.92	40.73	-73.94

Fig 16.0

Out[10]:

	price	number_of_reviews	availability_365	room_type	neighbourhood_group
cluster					
0	117.63	8.66	22.02	Entire home/apt	Brooklyn
1	235.95	2.15	281.96	Entire home/apt	Manhattan
2	110.83	98.31	151.71	Private room	Brooklyn
3	126.81	14.27	286.92	Entire home/apt	Manhattan

Fig 17.0

number_of_reviews	last_review	reviews_per_month	calculated_host_listings_count	availability_365	cluster	pca1	pca2	cluster_label
9	2018-10-19 00:00:00	0.21	6	365	3	-0.246237	1.028253	Mid-priced Low-Demand
45	2019-05-21 00:00:00	0.38	2	355	3	-0.047888	1.734136	Mid-priced Low-Demand
0	No Review	0.00	1	365	3	-0.311292	0.791652	Mid-priced Low-Demand
270	2019-07-05 00:00:00	4.64	1	194	2	4.924250	1.380383	Budget High-Demand Rooms
9	2018-11-19 00:00:00	0.10	1	0	0	-0.641489	-0.920248	Mid-priced & Fully Booked

Fig 18.0

📍 Geographic Pattern Observations

- ➢ Manhattan dominates Clusters 1 & 3
  - Higher pricing power but lower demand variability

- ➢ Brooklyn dominates Clusters 0 & 2
  - Competitive pricing with stronger demand signals

🎯 Business Recommendations (Act Phase)

Cluster	Strategy Recommendation
Cluster 1 — Luxury Low-Demand	Introduce discounts, dynamic pricing, improve listing visibility
Cluster 3 — Mid-priced Low-Demand	Better amenities/marketing; optimize minimum-night settings
Cluster 0 — Fully Booked	Opportunity to slightly increase price to maximize revenue
Cluster 2 — High-Demand Budget	Expand private room offerings; upsell add-on services

Phase V - Share :

Dashboard :

- ➢ There are 46K total listings
- ➢ The average price is around $120/night
- ➢ Listings receive 24 reviews on average, totaling 1 million reviews
- ➢ Around 35K distinct hosts operate in NYC
- ➢ Manhattan is the most expensive area with an average price of $146
- ➢ Entire homes/apartments dominate the market (≈ 54% of listings)
- ➢ Private rooms are the second most common (≈ 26%)
- ➢ Shared rooms make up a very small share (≈ 20%)
- ➢ Brooklyn and Manhattan lead in number of reviews and listing volume
- ➢ Median price = $100
- ➢ Occupancy ≈ 70% (around 256 booked days per year on average)
- ➢ A geographic map highlights that listings are densely concentrated in major NYC boroughs

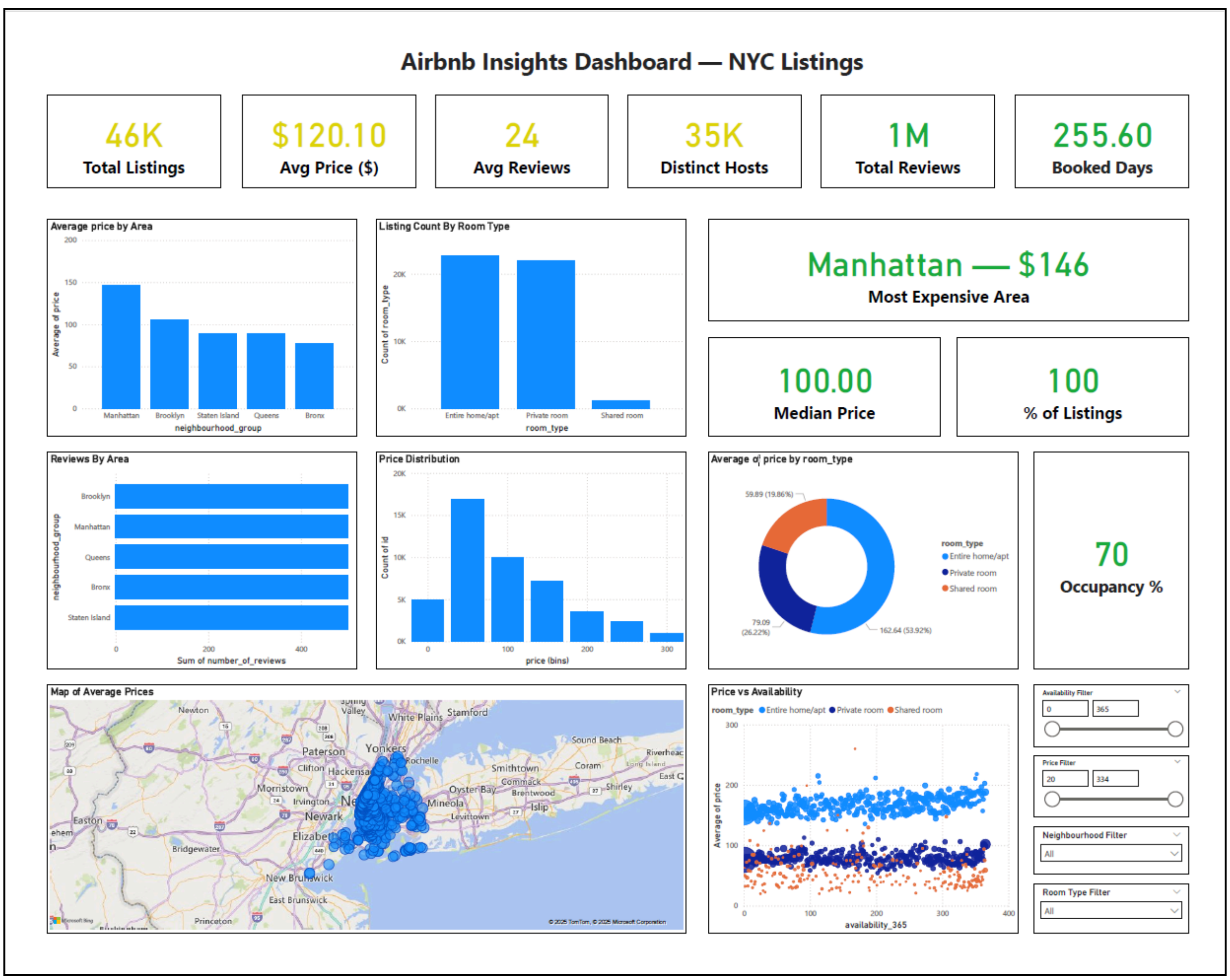

Fig 15.0

Phase VI - Act :

- **Target High-Demand Neighbourhoods**
  - Manhattan and Brooklyn have the highest average prices and reviews.
  - Expanding or optimizing listings in these boroughs can maximize revenue.
- **Focus on Entire Home/Apt Listings**
  - This category dominates the market and achieves higher pricing.
  - Hosts can increase profitability by improving full-space offerings.
- **Improve Performance in Low-priced Areas**
  - Bronx and Queens show lower pricing but strong review counts.
  - Marketing and listing quality improvements can help boost pricing competitiveness.
- **Enhance Listing Popularity**
  - Listings with higher review frequency attract more guests.
  - Encourage guests to leave reviews through better service and follow-ups.
- **Monitor Availability Strategy**
  - Very high availability does not guarantee higher bookings.
  - Hosts can adjust pricing dynamically during peak seasons to ensure better occupancy.
- **Leverage Data-Driven Pricing Models**
  - Pricing is influenced more by **location and room type** than availability or length of stay.
  - Implementing smart pricing tools can optimize revenue.

Conclusion :

Airbnb rentals in New York City show a highly active and competitive market, driven by the city's global tourism appeal. Pricing is largely influenced by the **neighbourhood and room type**, with central boroughs commanding premium rates. The strong presence of entire home rentals highlights the trend toward private and flexible travel experiences. Overall, Airbnb continues to be a major accommodation choice in NYC with healthy demand and significant economic impact in key boroughs like Manhattan and Brooklyn.